

# AI-Powered Contract Performance Guarantee Management System

An Intelligent Risk Assessment and Automated Lifecycle Platform Engineered to Support Strategic Infrastructure Contracts.

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PREPARED BY

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ORGANIZATION

**POWERGRID**

PROJECT TYPE

**AI-Driven Risk Intelligence**



## Current Challenges

**Manual CPG Tracking:** Laborious entry tables and spreadsheet dependency increase tracking overhead.

**Difficult Expiry Auditing:** Missing automated watchlists creates operational risks near final expirations.

**Delayed Alert Workflows:** Lack of automated notices means project teams operate without critical triggers.

**No Predictive Frameworks:** Absolute zero intelligence to measure real contractor delinquency risk.



## Negative Business Impact

**Missed Expiry Deadlines:** Inability to claim financial guarantees in default cases leads to unrecoverable claims.

**Exposure of Critical Assets:** Key power-grid systems and transmission pipelines are exposed to contractor stops.

**Heavy Manual Admin Overhead:** Engineer work hours are lost manually compiling tables and calling banks.

# Proposed Solution

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## Contract Hub

Centralized repository tracking profile scopes, financials, and major stakeholders.



## CPG Lifecycle

Automated pipeline mapping guarantee assets from initial award to final closeout.



## Cloud Repository

Reliable cloud attachment tracking supported by custom Cloudinary blob integrations.



## Risk Pipeline

Dynamic score engines identifying transactional anomalies and delivery delays early.



## AI Conversational Bot

A natural conversational assistant giving immediate answers to complex contract queries.



## Insights Engine

Actionable managers' dashboard showcasing operational alerts and performance statistics.

# Four Core Pillars

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PILLAR 01

## Contract Core

Structures core contractor credentials, project locations, and structural financials.



PILLAR 02

## CPG Lifecycle

Tracks active verification, active tracking, extensions, and closeout releases.



PILLAR 03

## AI Intelligence

Synthesizes health metrics, identifies anomalies, and outputs ML risk probability.



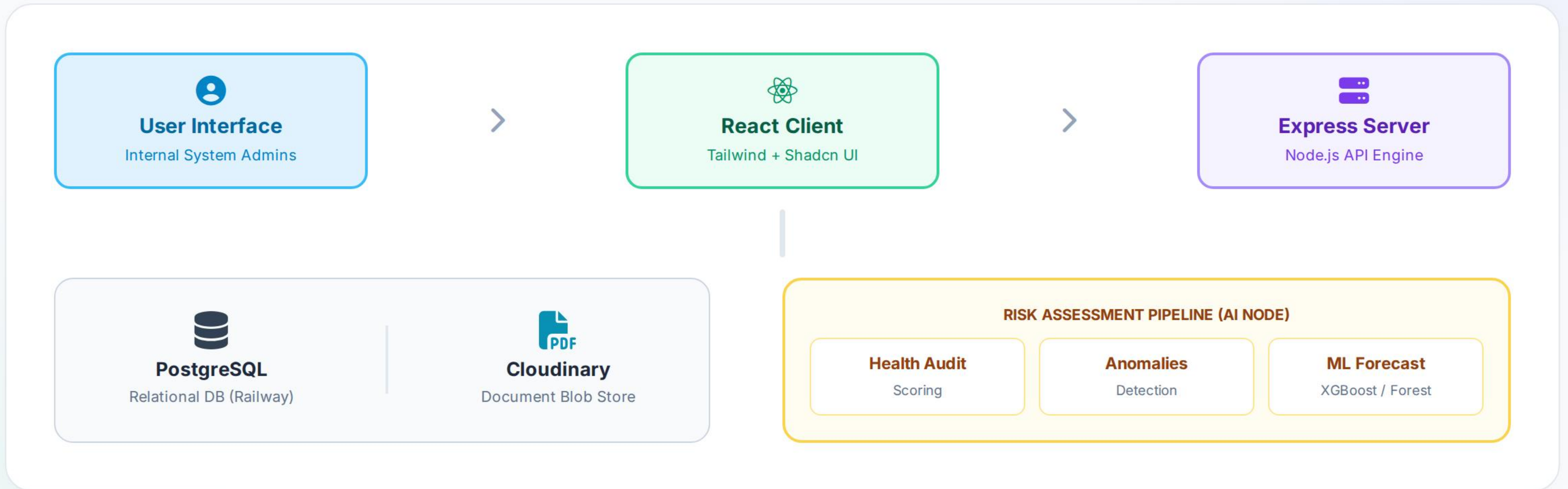
PILLAR 04

## AI Chatbot

Gives conversational access to complex queries and risk logs dynamically.

# System Architecture

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# Contract Creation Workflow

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## ☰ Action Sequence

1. Direct Security Authentication



2. Create Comprehensive Contractor Profile



3. Define Specific Contract & Value Parameters

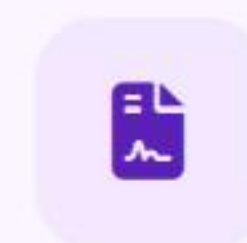


4. Upload Critical Guarantees Documents



5. Write Unified Record to PostgreSQL DB

## 🗄 Database Information Tracked



### Contract Information

Core identifier keys, signed names, milestones, active calendars.



### Financing and Security Caps

Total investment capital, designated bank lines, security percentages.



### Contractor & Verification Gaps

Profile risk factors, performance logs, delay instances.

# CPG Lifecycle Workflow

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The sequential lifecycle states mapping guarantees from initial execution to successful closeout or claim triggers.





## Trigger Events

The AI Engine monitors actions and initializes evaluation loops upon detecting:

- ✓ **Creation of a New Contract Record**
- ✓ **New CPG Upload & Verification Step**
- ✓ **Initiation of Extension Requests**
- ✓ **Direct Document Changes or status adjustments**



## Pipeline Assessment Run

The trigger fires an asynchronous worker script in Node Cron, scheduling analytics evaluations that run in the background to score contractor security indicators.

▶ **Running Background AI Jobs**

# Health Score Engine

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## Objective & Tracking Variables

Calculate a continuous health value (0-100) reflecting real-time performance profiles for active performance bonds.

### ACTIVE METRIC INPUTS

🕒 Days Remaining

↕ Extension Counts

📁 Missing Docs

💰 Contract Capital

## Scoring Formula

$$\text{Score} = 100 - R_{\text{Expiry}} - R_{\text{Extension}} - R_{\text{Delay}} - R_{\text{Compliance}}$$

**85 - 100**  
Healthy Status

**60 - 84**  
Monitor/Watch

**0 - 59**  
Risky System

# Anomaly Detection Engine

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## Operational Auditing

The database monitoring script flags contracts deviating from standard compliance envelopes:

- CPG Outlier Gaps:** Unusual guarantee-to-value anomalies
- Extension Clusters:** Extreme sequence of extension operations
- Document Faults:** Key bank confirmation documents absent

## REAL-TIME SCENARIO DETECTION

Standard Contract Avg

**₹2 Crore**



Flagged Current Record

**₹8 Crore**



### Structural Anomaly Flagged

Pushed to administrator's priority dashboard review.

# Machine Learning Risk Prediction

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## Model Input Attributes

> Duration & Cost

> Project Region

> Active Extensions

> Contractor History

> Historical Delays

> Expiry Distance

### CLASSIFICATION ALGORITHMS

Random Forest

XGBoost Classifier

### MODEL OUTPUT CASE INFERENCE

Inquiry: **Will BG-101 Need Extension?**

**YES**

Predictive Probability: **78% Score**

# Priority Ranking Engine

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### SORTING EVALUATION FORMULA

$$\text{Priority Score} = \text{ML Risk} + W_{\text{Health}} + W_{\text{Anomaly}}$$

### Case Instance Audit: BG-101

ML Risk: 85%

Health Score: 42

Anomaly: YES

Rank Placement: #1 Critical Urgency Alert

| Rank ID    | Guarantee Key | Health Score | ML Risk Probability | Mandated Operational Action |
|------------|---------------|--------------|---------------------|-----------------------------|
| #1 Urgent  | BG-101        | 42 (Risky)   | 85% High risk       | Execute Contract Call       |
| #2 Warning | BG-304        | 51 (Risky)   | 78% High risk       | Verify Documents Attach     |

# AI Assistant Interface

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CPG Chatbot Assistant

● Online

"Which performance guarantees are expiring in Odisha next month?"

"Scanning database... 3 high-value contracts flagged in Odisha.  
Exporting details spreadsheet for you."

Ask assistant something...



## ⚡ Intent Processing Pipeline

1. User natural language prompt query received



2. AI model runs intent and entity extraction



3. Direct dynamic database query translation & run



4. Conversational output synthesis & return

# Dashboard Design Showcase

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148

TOTAL CONTRACTS

92

ACTIVE CPGS

12

EXPIRING SOON

34

RELEASED SAFE

## ⚠️ Dynamic Risk Insights

17 Critical Active CPGs flagged for urgency metrics.

4 System Anomalies flagged during background audit checks.

## State Distribution

- Healthy: 55%
- Watch: 30%
- Risky: 15%



# Technology Stack Applied

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## Frontend

React JS

TypeScript

Tailwind CSS

Shadcn UI



## Backend

Node.js / Express

Prisma ORM

Node Cron Jobs

Nodemailer API



## DB & Cloud

PostgreSQL

Railway Hosting

Cloudinary API



## Artificial Intel

Gemini API Suite

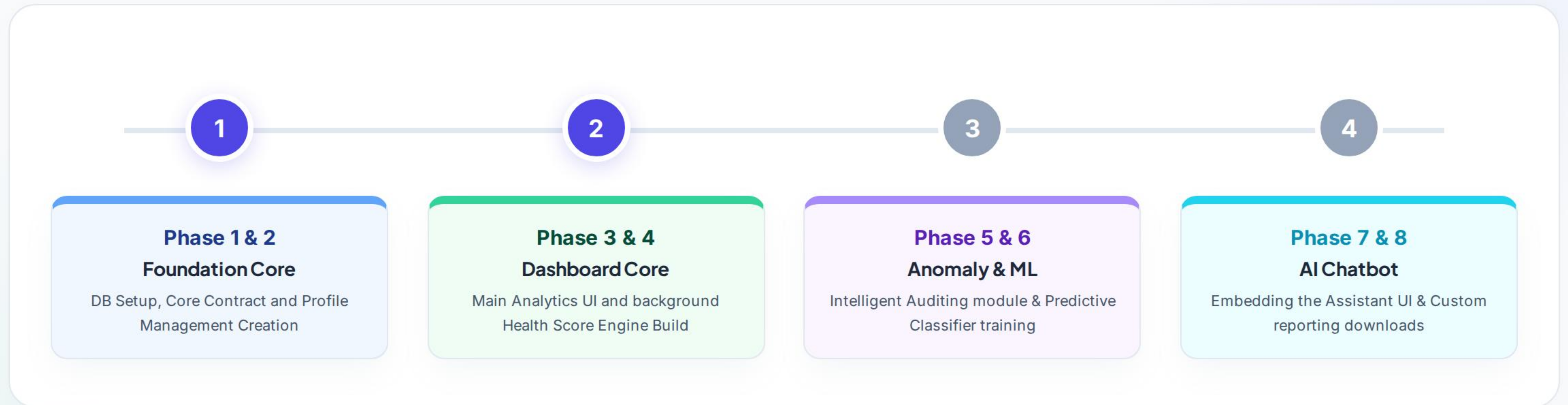
Python Data ML

Pandas / Numpy

Scikit-Learn models

# Development Roadmap

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# Expected Outcomes

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## Operational Benefits

- ✓ **Automated CPG Expiry Alerts:** Absolute zero human tracking slips.
- ✓ **Proactive Anomaly Detection:** Highlight extreme values early to defend audit tracks.
- ✓ **AI-based Decision Support:** Accurate risk probability forecasting before expiration.
- ✓ **Reduction of Administrative Work:** Direct storage, eliminating manual files.



## Future Scope

- + **Scale Architecture:** Migrate microservices setup backed by Kubernetes orchestrations.
- + **Stream Processing:** Employ Apache Kafka topics to broadcast system notifications instantly.
- + **Advanced Assistant (RAG):** Let the chatbot directly read through bank rules and clauses in pdf format.